大作业 2 掌纹识别身份确认要求

1）**Gabor**相**位**方法掌纹识**别**

2）**卷**积神经网络方法掌纹识**别**

3）**完**成实验报**告和**提**交代**码

4）**Deadline**第**15周**末/至**多2人一**组

卷积网络代码：https://github.com/pubozhang/mnist-from-scratch-to-pytorch

Filtering for Efficient Features

For tasks such as recognition, tracking, and segmentation, experience shows:

- With the “right” features, all algorithms will work well.
- With the “wrong” features, “good” algorithms will work marginally better than “bad/simple” algorithms, but it won’t work well.

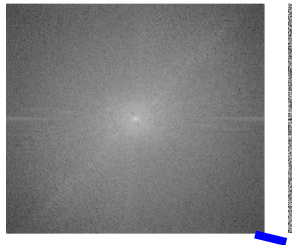
Phase Caries More Information

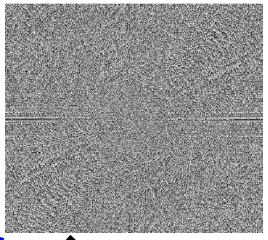
Raw Images:

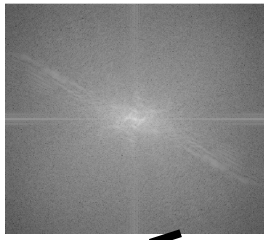


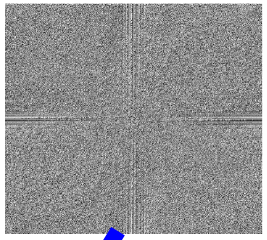


Magnitude and Phase:



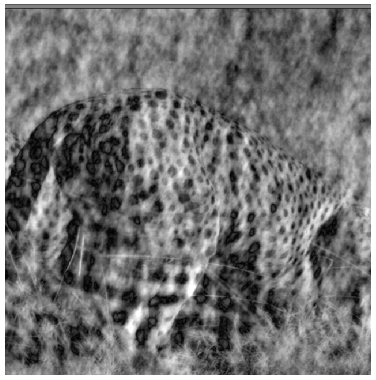


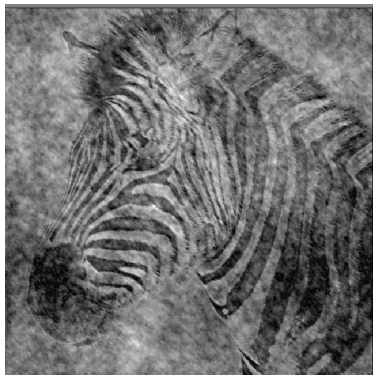




Reconstruct (inverse FFT) mixing the magnitude and phase images

Phase “Wins”

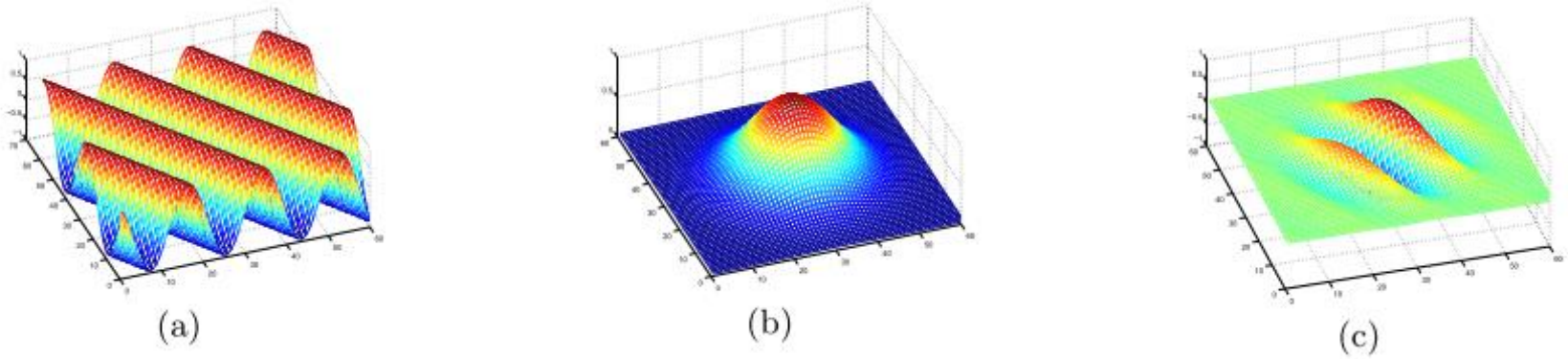




Gabor filters

- **Gabor filter**, named after Dennis Gabor, is a linear filter used for texture analysis, which means that it basically analyses whether there are any specific frequency content in the image in specific directions in a localized region around the point or region of analysis.
- Frequency and orientation representations of Gabor filters are claimed by many contemporary vision scientists to be similar to those of the human visual system, though there is no empirical evidence and no functional rationale to support the idea.
- They have been found to be particularly appropriate for texture representation and discrimination. In the spatial domain, a 2D Gabor filter is a Gaussian kernel function modulated by a sinusoidal plane wave.

Gabor filters

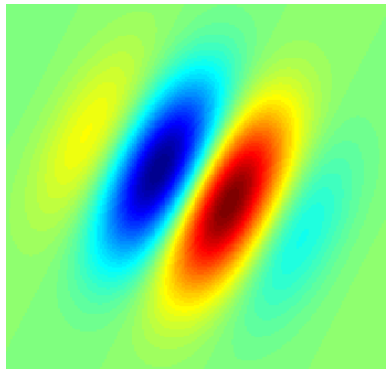


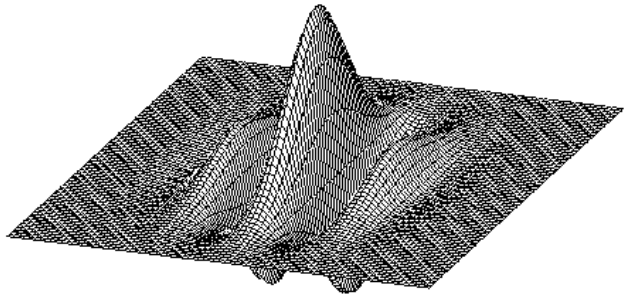
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- From uncertainty theory aspect, it reaches the lowest bound.

Gabor filters

Gabor filters became popular in vision partly because J.G Daugman (1980, ‘88, ‘90) showed that the receptive fields of most orientation receptive neurons in the (cat’s) brain looked very much like Gabor functions.

As with Gabor filters, the brain often makes use of over complete, non-orthogonal functions.





J.G.Daugman, “Two dimensional spectral analysis of cortical receptive field profiles,” Vision Res., vol.20.pp.847-856.1980

J. Daugman, “Complete discrete 2-d gabor transforms by neural network for image analysis and compression,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 36, no. 7, pp. 1169–1179, 1988.

Daugman, J.G. (1990) An information–theoretic view of analogue representation in striate cortex, *Computational Neuroscience*, Ed. Schwartz, E. L., Cambridge, MA: MIT Press, 403–424.

Gabor filters

$$g(x,y) = \frac{1}{2\pi\sigma_x\sigma_y}\exp[-\frac{1}{2}(\frac{\tilde{x}^2}{\sigma_x^2}+\frac{\tilde{y}^2}{\sigma_y^2})]\exp[2\pi jW\tilde{x}] \quad (1)$$

Gaussian Sinusoid

$$\begin{cases} \tilde{x} = x\cos\theta + y sin\theta \\ \tilde{y} = -x\sin\theta + y cos\theta \end{cases} \quad (2)$$

where σ_x^2 and σ_x^2 control the spatial extent of the filter, θ is the orientation of the filter and W is the radial frequency of the sinusoid.

The Fourier transform of the Gabor function in Eq. 1 is given by:

$$G(u,v) = \exp[-\frac{1}{2}(\frac{(u-W)^2}{\sigma_u^2}+\frac{v^2}{\sigma_v^2})] \quad (3)$$

where $\sigma_u = \frac{1}{2\pi\sigma_x}$, $\sigma_v = \frac{1}{2\pi\sigma_y}$.

Depending on one's task (object ID, texture analysis, tracking,…) one must then decide what size filters, in what orientations and what frequencies to use.

Gabor filters

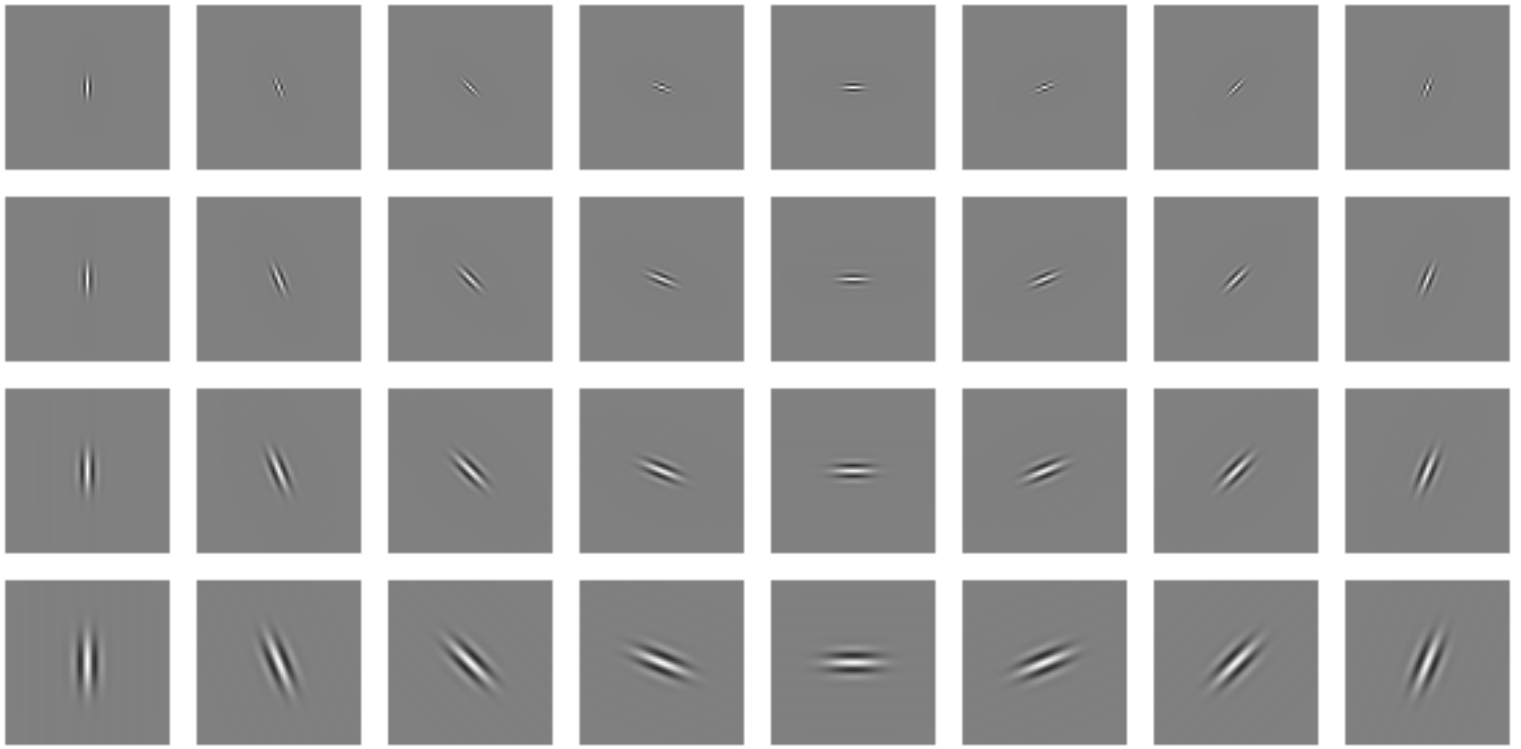
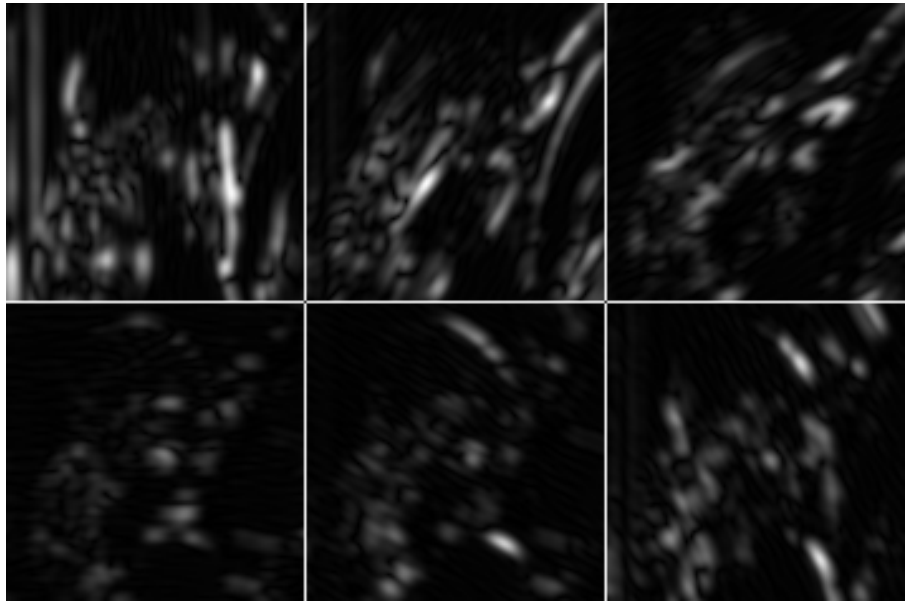


Image from Laurenz Wiskott, http://itb.biologie.hu-berlin.de/~wiskott/

Gabor filters





Gabor Magnitude and Phase

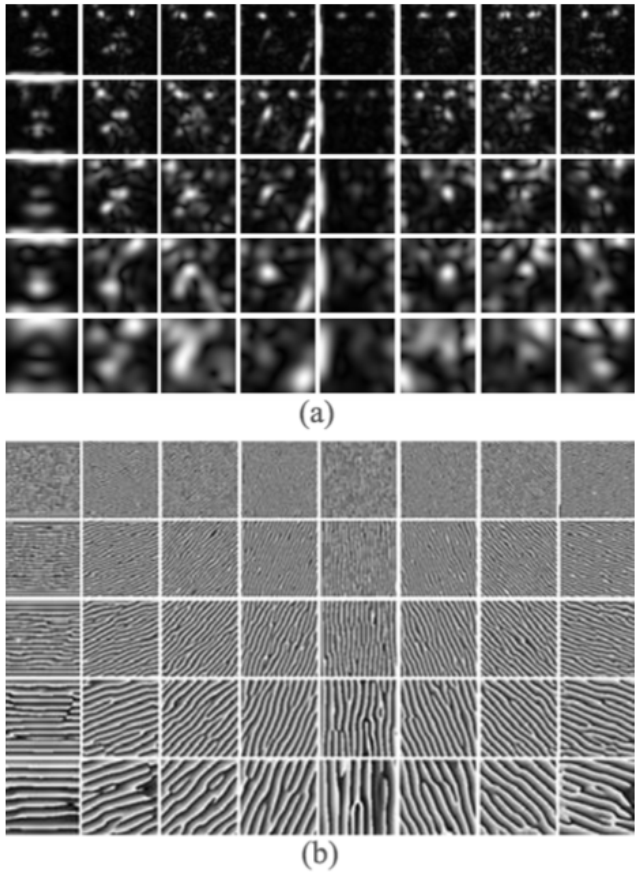
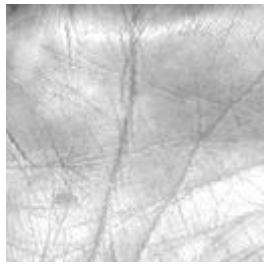
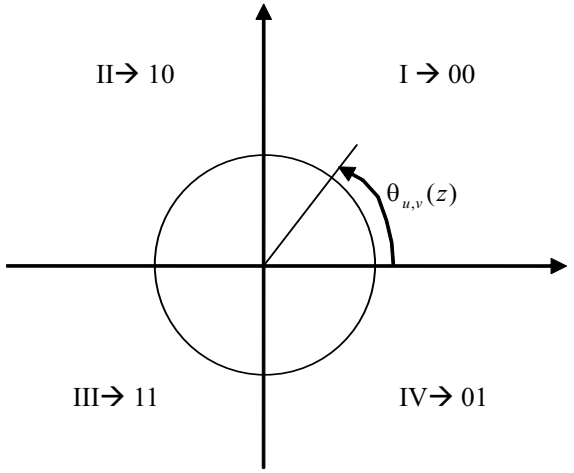


Fig. 1. Visualization of the Gabor phase and magnitude. (a) Gabor magnitude. (b) Gabor phase.

Gabor Quadrant Recognition

- Gabor Quadrant is successfully used on Iris and Palmprint recognition: Best Performance



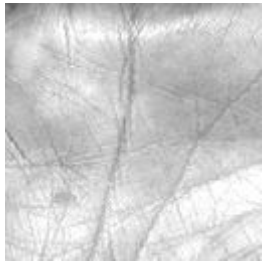


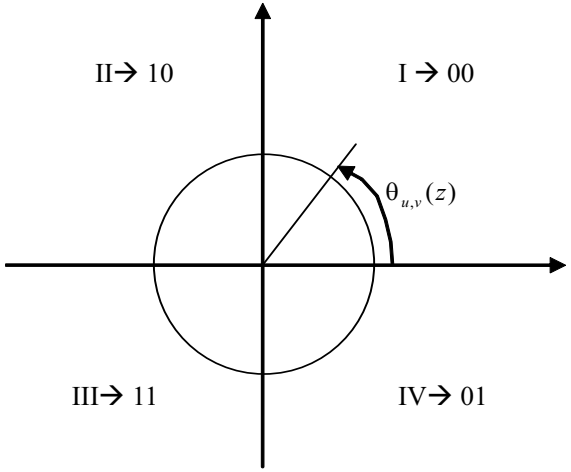


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- Gabor Quadrant is successfully used on Iris and Palmprint recognition: Best Performance
- D. Zhang, W.K. Kong, J. You and M. Wong, 2003, “On-line palmprint identification”, IEEE Trans. on Pattern Analysis and Machine Intelligence (PAMI) 25, 9, 1041-1050.

https://www4.comp.polyu.edu.hk/~biometrics/MultispectralPalmprint/MSP.htm

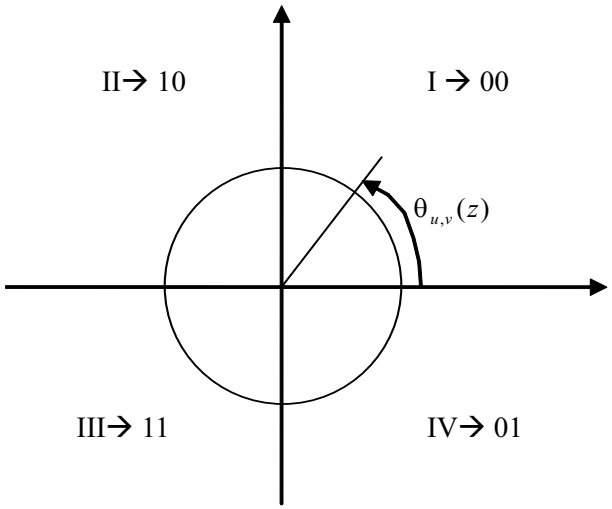






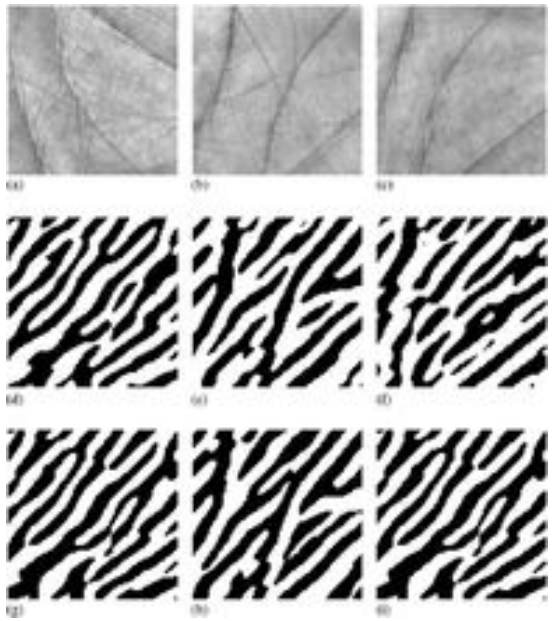
Gabor Quadrant

- Gabor Quadrant Feature



$$P^{\mathrm{Re}}_{u,v}(Z) = \begin{Bmatrix} 0, \text{if } \mathrm{Re}(G_{u,v}(Z)) > 0 \\ 1, \text{if } \mathrm{Re}(G_{u,v}(Z) <= 0 \end{Bmatrix} \quad P^{\mathrm{Im}}_{u,v}(Z) = \begin{Bmatrix} 0, \text{if } \mathrm{Im}(G_{u,v}(Z)) > 0 \\ 1, \text{if } \mathrm{Im}(G_{u,v}(Z)) <= 0 \end{Bmatrix}$$

$$P^{\mathrm{Re}}_{u,v}(Z) = \begin{Bmatrix} 0, \text{if } \grave{e}_{u,v}(Z) \in \{\mathrm{I}, \mathrm{IV}\} \\ 1, \text{if } \grave{e}_{u,v}(Z) \in \{\mathrm{II}, \mathrm{III}\} \end{Bmatrix} \quad P^{\mathrm{Im}}_{u,v}(Z) = \begin{Bmatrix} 0, \text{if } \grave{e}_{u,v}(Z) \in \{\mathrm{I}, \mathrm{II}\} \\ 1, \text{if } \grave{e}_{u,v}(Z) \in \{\mathrm{III}, \mathrm{IV}\} \end{Bmatrix}$$



Palmprint identification

A demo of matching: Hamming Distance

1	2	2	1
1	2	1	1
4	0	2	2
1	-2	1	-2

1	2	2	1	0
1	2	1	1	0
4	0	2	2	0
1	-2	1	-2	0

0	0	0	0
0	1	3	0
0	2	2	0
0	0	0	0

0	1	2	2	1
0	1	2	1	1
0	4	0	2	2
0	1	-2	1	-2